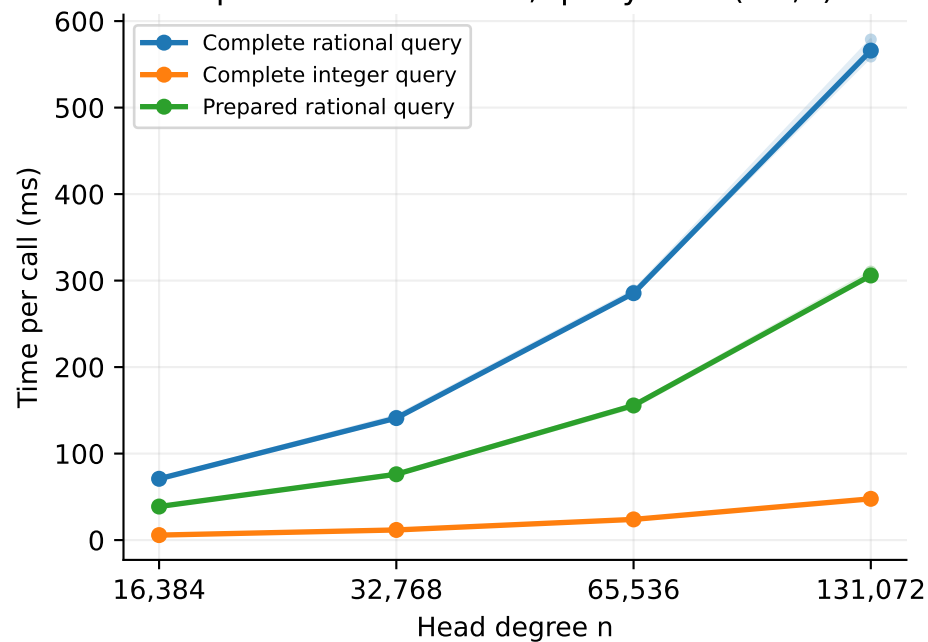
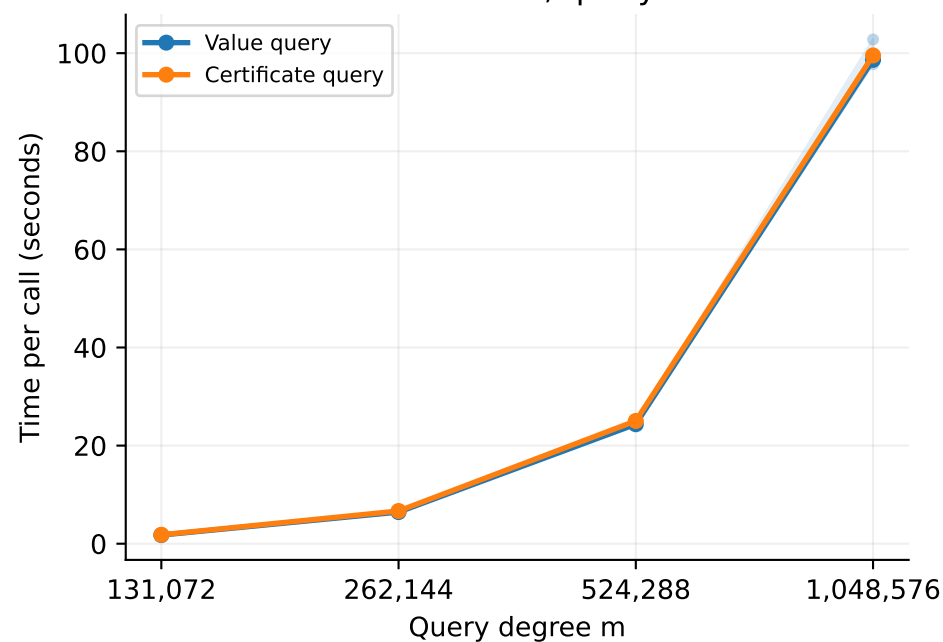


Sturm: elapsed time and memory versus polynomial size

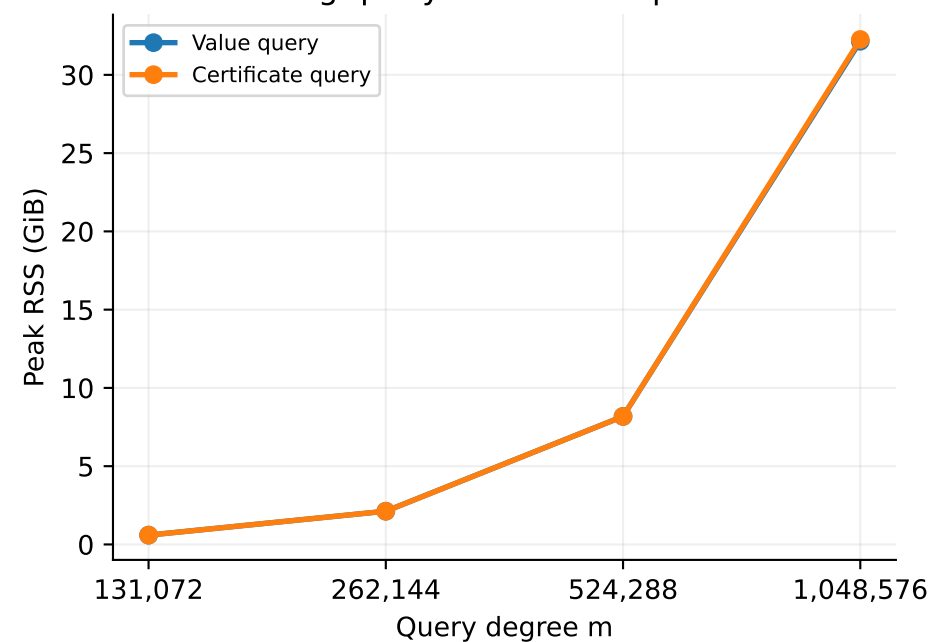
Sparse head $2X^n - 1$, query 1 on $(-1,1)$



Fixed head $X^2 - 2$, query $X^m + 1$



Growing query: whole-child peak RSS



Shared host chungus2, AMD EPYC 9455; 4 trials per rung. Dots: every trial; shaded area: min-max.

Retained source scopes: c5efa57c (head degree), 4f44b806 (query degree). Prepared inputs excluded; whole-child RSS includes preparation. No fitted curve or budget line.